

On a road to nowhere: a folded universe can touch itself only in four large dimensions

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Abstract

If CPT holds of the universe and not only of its laws, spacetime is two sheets joined at the Big Bang. Where can a point reach its own image on the other sheet? Outside any horizon, nowhere: the two are spacelike separated and the cross-sheet commutator vanishes identically. Inside a black hole the answer turns on two distances: the half-turn round the sphere the fold demands, and how far round a falling path can actually get. Four large spacetime dimensions is the only case in which the second is the larger, and contact then fills the region inside half the horizon radius. Five dimensions fall short by exactly nothing. The region so selected carries closed timelike curves, and the image stress defocuses them.

Main

Every observer sits inside a horizon. A black hole has one, and so does the universe, because expansion carries distant things away faster than their light can return. The two have different causes and the same causal structure, and the same mathematics describes both.

Suppose that CPT, the combination of exchanging matter for antimatter, reflecting space and reversing the direction of time, is a property of the universe as a whole. Every experiment ever performed has respected CPT. Boyle, Finn and Turok proposed taking it of the universe as a whole [1,2] and it gives the Big Bang a far side, carrying a universe like ours with all three of those reversed. An involution relates the two sheets. We call it the fold, and the first thing to say about it is that it is an operation and not a place. There is no seam, no boundary and no direction you could travel in to find one. Every point of our sheet simply has a partner on the other.

The map is not new. Identifying antipodal points of the extended Schwarzschild manifold is Gibbons' elliptic interpretation [3], written in the form used here by Sanchez and Whiting [4], and gauging a spacetime inversion of this kind is treated by Harlow and Numasawa [5]. What follows takes the fold as given; what it implies for the field equations and for the matter sector is set out elsewhere [6,7]. We have not found the contact radius below stated in any of that work, and claim it as new on that basis rather than on a search we can call complete.

Where, if anywhere, can a point reach that partner?

By *reach*, and by *touch* below, we mean that the commutator between a field at a point and the same field at its fold image stops vanishing, so the two can influence one another. Nothing travels between the sheets in what follows. No matter crosses, and no traversable connection is predicted anywhere.

Outside a horizon the answer is nowhere, and that is a theorem

On the cosmological geometry the fold implies, every point is spacelike separated from its own image, and the proof is one line. Write de Sitter space as the hyperboloid $X \cdot X = \ell^2$ in five-dimensional Minkowski space and let $Z(X, Y) = X \cdot Y / \ell^2$, which equals one when two points coincide and exceeds one when they are timelike separated. The fold sends X to $-X$, so

$$Z(X, -X) = -1,$$

which is as far into the spacelike range as the invariant reaches. The commutator therefore vanishes identically at every such pair, and the two sheets are correlated everywhere while able to signal nowhere.

Geometrically, de Sitter space is conformal to a piece of a cylinder whose spatial slices are round three-spheres and whose conformal time runs over an interval of length π . The fold is the antipodal map of the sphere together with a reflection of that time, and antipodal points on a sphere are π apart wherever they sit. So a light ray leaving any point runs out of conformal time before it has covered half the way round, by a margin that closes only at infinity. The companion works this out in full [7].

That silence is not a limitation of instruments and it will not yield to a better experiment. It is the same rule that stops anything outrunning light.

A black hole's exterior needs its own argument and gets a shorter one. There the fold acts as the wedge reflection, which scales the Kruskal coordinates as $dU = 2U$ and $dV = 2V$, so $dU dV = 4UV$ and the radial part of the separation turns timelike exactly where $UV > 0$. That region is the interior and nothing else, so the commutator is zero outside the horizon and nonzero inside, changing over at the horizon because UV changes sign there and not because anything was tuned to put it there. That is where to look next.

Inside a black hole, two distances compete

The fold acts on the transverse sphere as the antipodal map, sending each direction to its opposite. Your image therefore lies half a world away, in the exact sense that the south pole lies half a world from the north: whichever way you set out, you must go a quarter of the way round and then a quarter again. That is half a turn, π of angle, and it never changes. It does not depend on the mass of the hole, and it does not depend on the number of dimensions, because it is fixed by the antipodal map being a free involution of a sphere and by nothing else.

How far round you can actually get is the other half of the question, and falling is what limits it. Inside a Schwarzschild horizon the radial part of the separation between a point and its image turns timelike, so as far as the radial direction is concerned a connecting curve exists. Sideways is where the difficulty is. Requiring the path to be causal caps the angle it can sweep, and for one crossing of a Schwarzschild interior in D dimensions, where $f(r) = 1 - (r_h/r)^n$ with r_h the horizon radius and $n = D - 3$, that cap is

$$\int_0^{r_h} \frac{dr}{r\sqrt{|f|}} = \frac{1}{n} \int_0^1 \frac{du}{\sqrt{u}\sqrt{1-u}} = \frac{\pi}{n},$$

on substituting $u = (r/r_h)^n$. The horizon radius cancels on the way, which is why the mass never enters, and the remaining integral is $B(\frac{1}{2}, \frac{1}{2})$.

A connecting path is allowed two such crossings and no more, one through the black-hole interior and one through the white-hole interior of the extended geometry. Figure 1 shows why there cannot be a third: nothing leaves a black hole and nothing re-enters a white one, so a curve running from an image to its point uses each interior once. Two crossings give $2\pi/(D-3)$, against the π that is needed.

In four large dimensions you can go twice as far as you need

The angle a path can still cover, starting from radius r , works out to

$$\Delta\phi_{\max}(r) = 2\pi - 4 \arcsin \sqrt{\frac{r}{2M}},$$

which equals the required π at $r = M$ and exceeds it everywhere inside. Contact therefore fills the inner half of a Schwarzschild interior, every point at or inside half the horizon radius, and nowhere else in that spacetime.

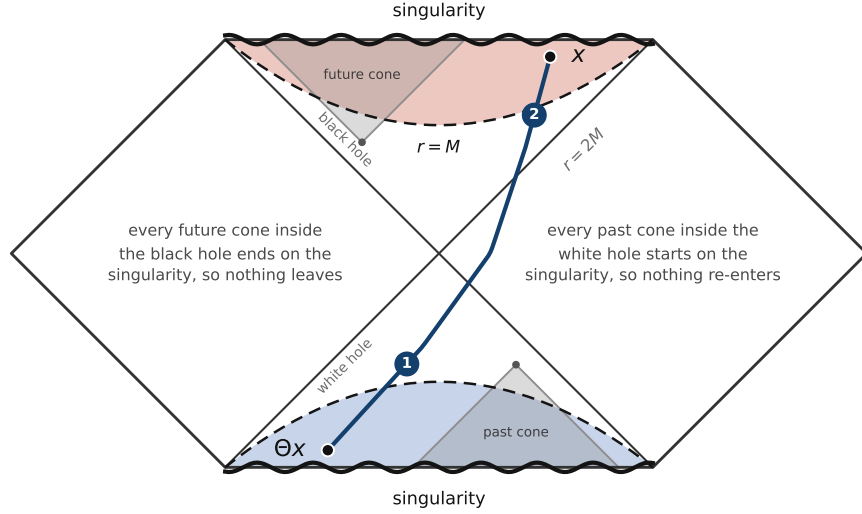


Figure 1: FIG. 1. The eternal Schwarzschild geometry, conformally compactified, with a point x and its fold image Θx . The fold acts here as reflection through the bifurcation surface, so the image of a point in the black-hole interior lies in the white-hole interior. The two grey cones are the reason a connecting path crosses an interior exactly twice, at 1 and 2: every future cone inside the black hole is capped by the singularity and reaches no horizon, and every past cone inside the white hole is capped the same way, so a causal path running from the image to the point enters each interior once and has no third crossing open to it. Shaded in colour is $r \leq M$, where the two sheets can touch. The fold reverses time orientation, so those two regions are exact time-reverses of one another and the colours say so: light escaping the black-hole side reaches an exterior redshifted, and the same factor reversed makes light falling toward the white-hole side arrive blueshifted. Null rays run at forty-five degrees everywhere and that structure is exact; the constant- r curves are drawn at eased heights, since in this conformal frame $r = M$ sits at 0.469 against the singularity at 0.5 and the contact region would otherwise be a sliver.

The distance is reached rather than approached. A null geodesic sitting at fixed Killing position and spiralling covers it exactly, so the statement is an equivalence: inside $r = M$ the sheets can reach each other, outside it they cannot.

The same comparison picks out four dimensions

Nothing above used $D = 4$. Setting how far you can go against how far you must go, four dimensions gives 2π against π , five gives π against π , and six and above give at most $2\pi/3$. Figure 2 has the comparison at every depth.

Five is the case to look at. There you can go exactly as far as you must and no further, so a path can approach its image without ever arriving, and the gap closes only in a limit no path attains. Six dimensions and above fall short at every depth. Four is the only case in which the two halves of a folded universe can reach each other at all, and it clears the distance twice over.

We were not looking for this. The angular bound was being calculated for another purpose and the dimension dependence fell out of it. Any argument that selects the dimension we happen to inhabit deserves suspicion. These two numbers have separate origins. The distance required comes from the topology of the antipodal map, the distance available from the causal structure of the interior. Neither knows about the other, and only their ratio selects anything.

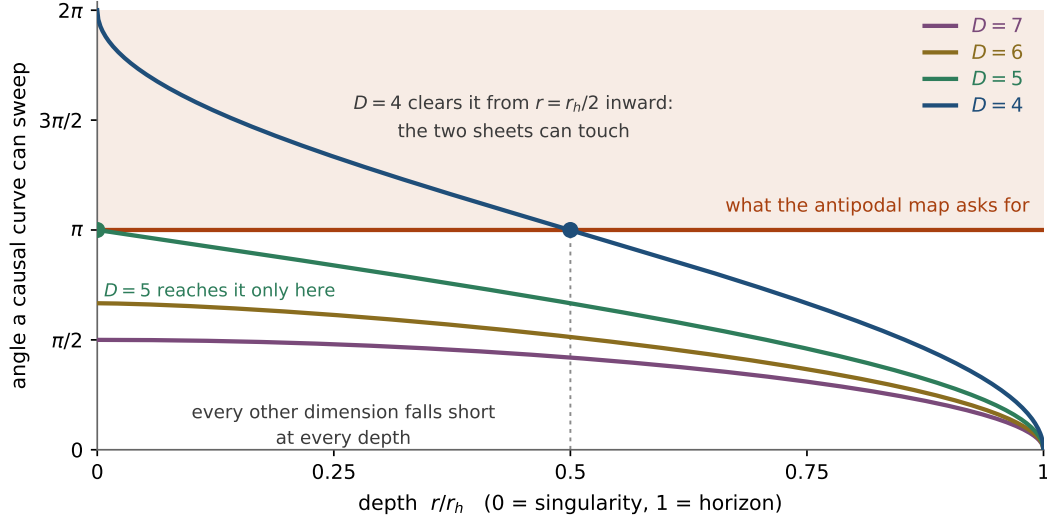


Figure 2: FIG. 2. The angle a causal curve can still sweep on its two crossings of an interior, against depth into a Schwarzschild interior, in four to seven spacetime dimensions. The horizontal line is the half turn the antipodal map asks for, which does not move with the dimension. Only $D = 4$ rises above it, and it does so from half the horizon radius inward. $D = 5$ meets the line at the singularity and nowhere else.

What the region turns out to be

The fold pairs every point with a distinct one, so it is a free involution and the quotient is a manifold on which a point and its image are the same point. On that quotient, a point able to influence its own image is a point with a closed causal curve through it.

So the inner half of the interior is the causality-violating core of the folded hole. Closed timelike curves lie strictly inside it, the closed null geodesic lies exactly on its boundary, nothing lies outside it, and the whole of it is sealed behind a horizon and then behind half of what remains.

Predicting closed timelike curves is normally a reason to abandon a model. Hawking's chronology protection conjecture [8] holds that the laws of physics prevent them from forming, and it has stood since 1992 without a general proof. What stands behind the conjecture is a theorem of Kay, Radzikowski and Wald: a Hadamard state's two-point function must be singular on the horizon, given a compactly generated Cauchy horizon on a time-orientable spacetime [9]. Neither hypothesis holds here. A Cauchy horizon is null, and the surface that bounds the region here is not: the closed null geodesic lies on it, but the surface itself is $r = M$, and surfaces of constant r are spacelike inside a hole. The fold also reverses time orientation, so the quotient is not time-orientable. That theorem therefore reaches no verdict on this case, in either direction.

The region works against itself

With no theorem either way, what settles the case is what the region does to itself. The connecting curves refocus at a caustic, and the null-null contraction of the image stress there comes out negative. Two routes give that sign and they share no assumption: moving the mass at conformal coupling, and moving the coupling at zero mass. Behind the first is a mode sum that agrees with the closed-form massless result on the same geometry to 6×10^{-4} , and whose contraction is negative at every mass and close to proportional to m^2 . The two routes approach the caustic at different rates, $\delta^{-4.010}$ for the coupling and $\delta^{-2.037}$ for the mass, so what they share is the sign and not a coefficient.

Whether the sign survives the move to a black hole is settled by calculating it. The same image two-point function is evaluated twice by methods with nothing in common, once as a mode sum carrying no Van Vleck

factor anywhere and once by geometric optics carrying one, and wherever both apply they agree. Any relative phase between them would have to be a multiple of π , since a Wightman function at spacelike separation is real, and they agree, so the phase is zero.

Schwarzschild is Ricci-flat, which closes the coupling channel and leaves the mass one, so the sign at the caustic holds for any field that is not massless. A negative null-null contraction defocuses. The image stress therefore acts against the very congruence that would bring the two sheets into contact, and the region undoes the structure that creates it. On this argument chronology protection holds here, for a reason that theorem cannot supply.

That is evidence for the conjecture and not a proof of it, and here is the gap. The negative sign is established at the caustic. Carrying it to the horizon that bounds the region rests on conservation of the image stress along the congruence, not on a mode sum evaluated inside the hole. That sum is the missing step.

What this does and does not claim

No test comes with this. The contact region lies inside the inner half of an event horizon, so it is unreachable in principle and nothing here can be checked by going to look.

Nothing above treats rotation. The fold's map carries to Kerr exactly, but the contact derivation does not: the angular sweep on a distorted sphere is its own calculation. A first pass in [7] gives a contact radius falling from $r = M$ at zero spin to about $0.89M$ at $a = 0.9$, so spin shrinks the region where charge left it alone.

Everything above is done on a Schwarzschild hole. Adding charge moves the boundary to $r \leq (r_+ + r_-)/2$, half way between the outer and inner horizons at every charge, so the result is about the inner half of the interior and half the horizon radius is the uncharged special case. Reaching that region is proved at every charge; ruling out everything outside it uses an angle-maximising curve that Schwarzschild has and a charged hole does not, so at charge the inner half is a floor on the contact region rather than the whole of it [7].

The second of the two crossings runs through a white-hole interior, so the result is about an eternal, two-sided hole. A hole formed by collapsing matter has no past singularity and no second exterior, and none of this reaches one.

Four large dimensions is the only case in which a fold of this kind permits contact. That is a property of the fold, not a derivation of the dimension of the world. Compactified extra dimensions are a separate case, worked in [7]: small ones are not excluded, and the fold's contact switches off below the compactification scale.

Take CPT to hold of the universe, not only of its laws. The two sheets then touch in the inner half of a Schwarzschild interior and nowhere else in that spacetime, in four large spacetime dimensions and in no other. Where they touch, the image stress defocuses the congruence that would have joined them. And the four is a ratio: an angle the antipodal map fixes by topology, against an angle the interior fixes by causality.

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